

**Can we write multiple methods with same name in a java class**

Yes, we can write multiple methods with same name but parameter must be **different** otherwise code will not compile. [Known as Method Overloading]

```
Case 1 :
-----
Method name same, Parameters are different so, It is valid
-----
void main()
{
    accept(5);
}

void accept(int x)
{
}

void accept(int x, int y)
{
}

Case 2 :
-----
Method name same, Parameters are parameters are also same so, It is invalid
-----
void main()
{
    accept(5);
}

void accept(int x)
{
}

void accept(int x)
{
}
```

Note: - We can also write multiple main methods (main method can be overloaded) with different parameter but JVM will always execute the main method which takes String [] args (String array) as a parameter as shown in the program below.

```
Case 1:
-----
void main ()
{
    IO.println ("Hello India");
}

public static void main(String [] args)
{
    IO.println("Hello World");
}

Output : Hello World

Case 2 :
-----
void main(String [] args)
{
    IO.println("Hello India");
}

void main()
{
    IO.println("Hello World");
}

Output : Hello India

Case 3 :
-----
void main(int [] args)
{
    IO.println("Hello India");
}

void main()
{
    IO.println("Hello World");
}

Output : Hello World

Case 4:
-----
void main(int [] args)
{
    IO.println("Hello India");
}

void main(String [] args)
{
    IO.println("Hello Java");
}

void main()
{
    IO.println("Hello World");
}

Output : Hello Java
```

So, final conclusion, Java provides the priority to the main() method using following order:

- a) public static void main(String [] args)
- b) void main(String [] args)
- c) void main()

Note: we cannot write a and b together due to ambiguity. Both are accepting String [] args as a parameter

### String [] args:

String is a predefined class and args is an array variable of type String so, it can hold multiple values.

IQ: Why the main method of java accepts String array as a parameter?

String is a collection of alpha-numeric character so it can accept all different kind of values. Java software people have provided String array as a parameter so it can ACCEPT **MULTIPLE VALUES OF DIFFERENT TYPE**, that means providing more wider scope to accept heterogeneous types of values.

### What is command line argument in java ?

\* In Java, Whenever we pass arguments to the main method then it is called "Command line Argument".

```
Case 1 :
-----
public void accept(int x) //x is parameter
{
}

Case 2 :
-----
void main(String [] args) //Command line argument
{
}
```

\* By using command line argument, We can pass some value at runtime during the execution of the program.



\* The advantages of command line argument is, Single time compilation and multiple times execution with different values.

//Program :

Command.java

```
void main(String []args)
{
    IO.println(args[0]);
}
```

javac Command.java

java Command Scott Smith

Output is : Scott

Command1.java

```
void main(String []args)
{
    IO.println(args[1]);
}
```

javac Command.java

java Command Scott Smith

Output is : Smith

Fruits.java

```
void main(String []fruits)
{
    for(String fruit : fruits)
    {
        IO.println(fruit);
    }
}
```

javac Fruits.java

java Fruits Apple Mango Orange

Output : Apple Mango Orange

Square.java

//Fins the square of a given number

```
void main(String [] args)
{
    int num = Integer.parseInt(args[0]);
    IO.println("Square of "+num+" is:"+ (num*num));
}
```

SortColorsInAscending.java

//Fins the square of a given number

```
void main(String [] colors)
{
    Arrays.sort(colors);
    for(String color : colors)
    {
        IO.println(color);
    }
}
```